Cyclistic Business Analysis Report

Introduction:

Cyclistic is a fictional bike-share company in Chicago. The director of marketing, Lily Moreno, believes the company's future success depends on maximising annual membership sales. Therefore, wants to understand how casual riders and annual members use Cyclistic bikes differently. From these insights, a new marketing strategy is to be designed to convert casual riders into annual members.

Until now, Cyclistic's marketing strategy has relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the flexibility of its pricing plans: **single-ride passes**, **full-day passes**, and **annual memberships**. Customers who purchase single-ride or full-day passes are referred to as **casual riders**. Customers who purchase annual memberships are **Cyclistic members**.

Business Task:

The director of marketing needs to better understand how annual members and casual riders differ, why casual riders would buy a membership, and how digital media could affect their marketing tactics.
I am tasked with analyzing the Cyclistic historical bike trip data to identify trends, present insights, and make recommendations.

Data Description:

This dataset comprises 12 months of ride data from the company's database for the year 2025. Each record contains the following fields:

- **ride_id:** The primary key uniquely identifying each ride.
- **rideable_type:** The type of bicycle used for the trip (e.g., electric or classic).
- **started_at:** The date and time the bike was undocked (start of the ride).
- **ended_at:** The date and time the bike was docked (end of the ride).
- **start_station_name:** The address of the station where the bike was undocked.
- **start_station_id:** The unique identifier for the starting station.
- **end_station_name:** The address of the station where the bike was docked.
- **end_station_id:** The unique identifier for the ending station.
- **start_lat:** The latitude of the starting station.
- **start_lng:** The longitude of the starting station.

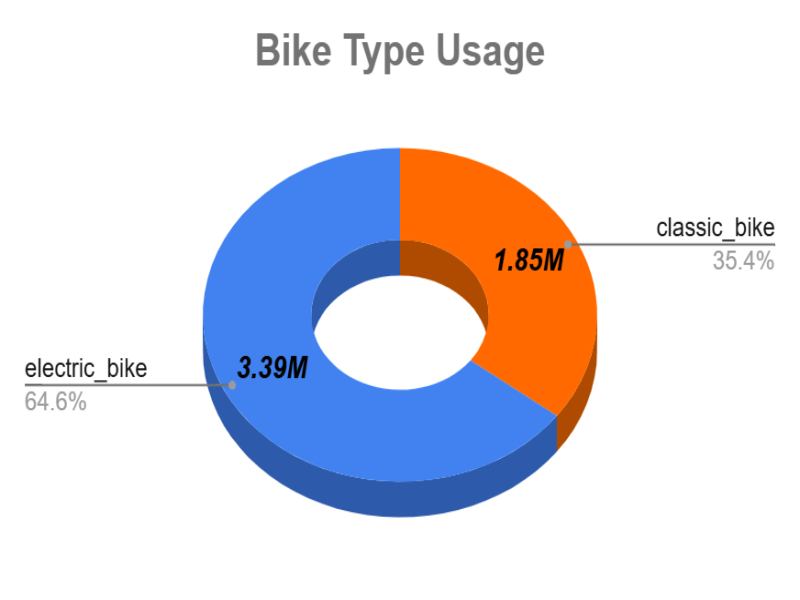
- **end_lat:** The latitude of the ending station.
- **end_lng:** The longitude of the ending station.
- **member_casual:** The type of pass used for the trip (casual rider or annual member).

Data Processing Documentation:

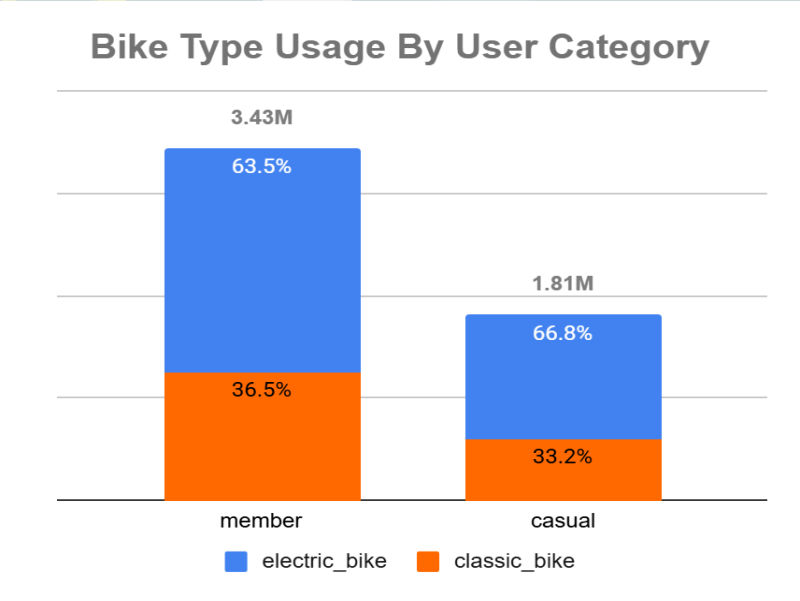
1. The 12 months of data were uploaded to a SQL database using **BigQuery** and merged to create a single annual dataset.
2. Preprocessing checks were performed on the data, revealing the following:
    - All ride IDs were unique, indicating no duplicate entries
    - The company utilizes two bike types: electric and classic.
    - Station names and IDs had some missing values. These records were retained as they accounted for approximately ***33.5%*** of the total data.
    - Some trip start times were recorded as being later than the end times, which indicates a data entry error.
3. A new column, **`ride_length`**, was calculated as the difference between a trip's start and end times.
4. The data was filtered to include only trips with a ***ride_length*** between **1.5 minutes** and **1 hour**. This step was taken to exclude false starts (*<1.5 minutes*) and unusually long trips (*>60 minutes*), which may indicate a rider forgot to dock the bike.

Analysis Results:

Twelve months of 2025 data were merged using the **`UNION ALL`** statement in BigQuery, resulting in one table with ***approximately 5.55 million rows*** of total annual data. This table was then exported to Google's Connected Sheets for analysis in a familiar spreadsheet environment. The data was filtered to include only trips with a *ride_length* between 1.5 minutes and 1 hour, reducing the observation count to ***approximately 5.24 million rows***. The key findings from the analysis are as follows:

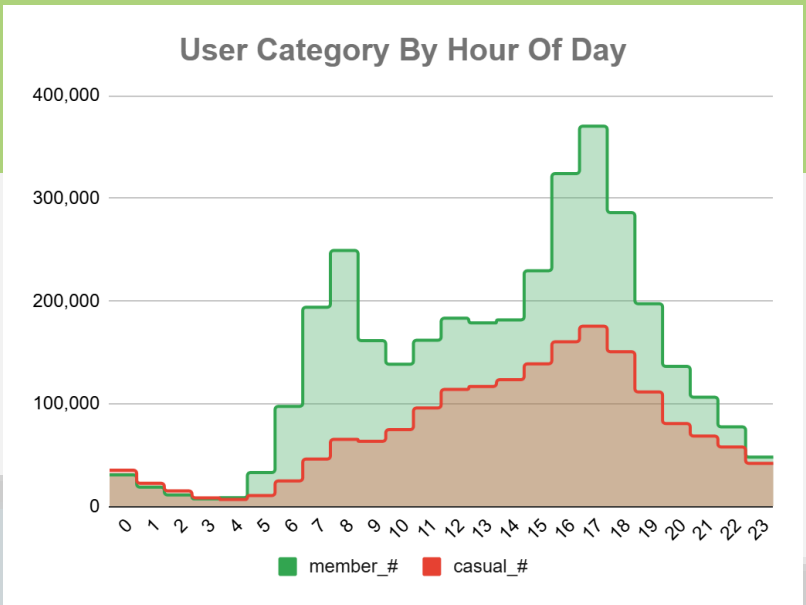


- **Bike Type Preference:** Of the total rides (***~5.24M***), **64.6%** were taken with **electric bikes**, compared to **35.4%** with **classic bikes**, indicating a strong overall preference for electric bikes.

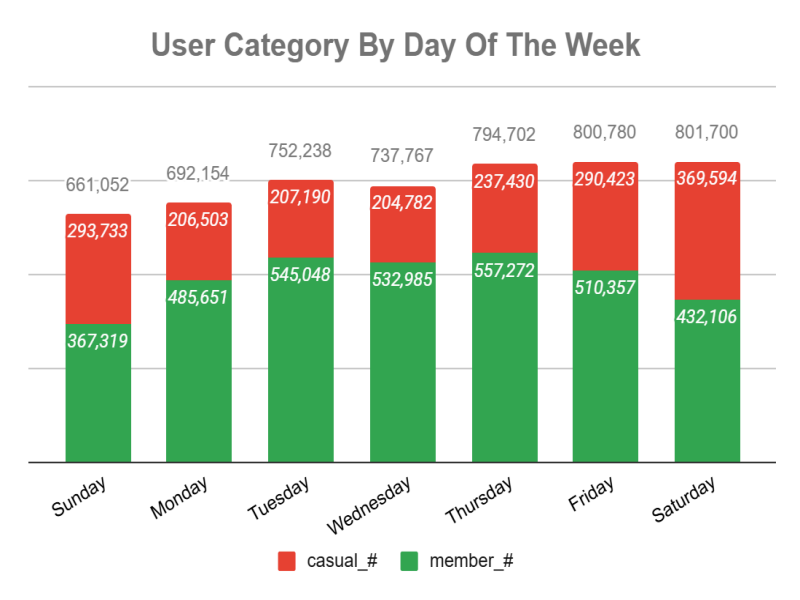


- **Bike Preference by User Type:** This preference for electric bikes holds true across both user categories:
  - **Members (~3.43M trips):** ***63.5%*** used electric bikes versus ***36.5%*** using classic bikes.
  - **Casual Users (~1.81M trips):** ***66.8%*** chose electric bikes versus ***33.2%*** using classic bikes.

This suggests that both user types generally prefer electric bikes for their trips.

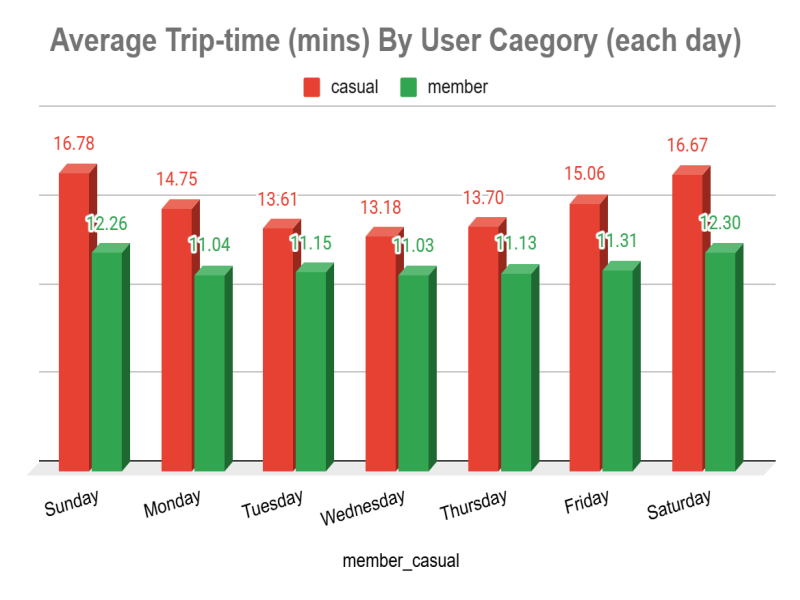


- **Preferred Trip-Start Hours:** A deeper analysis of ***trip-start hours*** revealed distinct usage patterns:
  - **Members** showed a **bi-modal** distribution, peaking in the early morning and evening. This pattern supports the assumption that members primarily use the service for work commuting.
  - **Casual Users** displayed a **uni-modal** distribution, peaking in the evening (leisure hours). This difference highlights a contrast in usage purpose, likely distinguishing work commute from leisure.

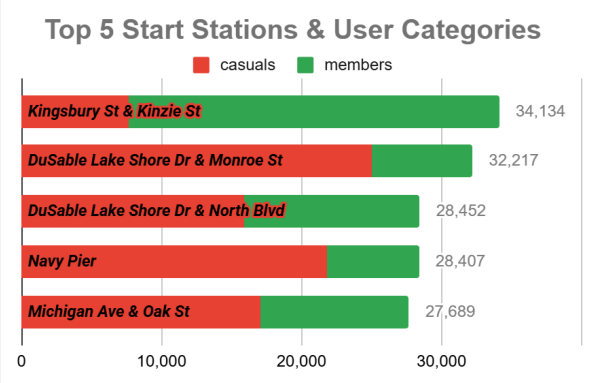


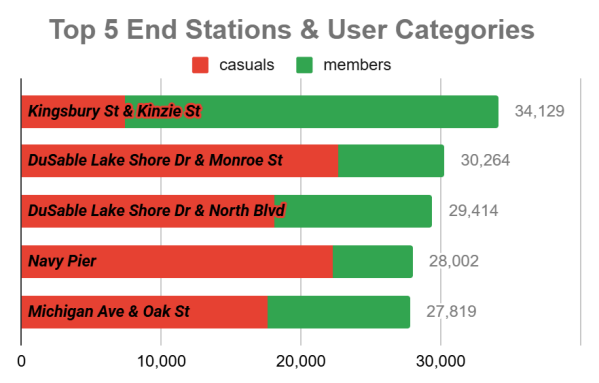
- **Weekly Usage Patterns:** Analysis of ***ride counts*** across the week showed a significant discrepancy between the two categories:
  - **Member** usage remained consistently high throughout the weekdays but dropped sharply on weekends.
  - **Casual User** usage was low on weekdays but rose spontaneously on weekends, peaking on Saturdays.

    This reinforces the suggested mindset shift between the two groups: ***work-focused*** versus ***leisure-focused***.

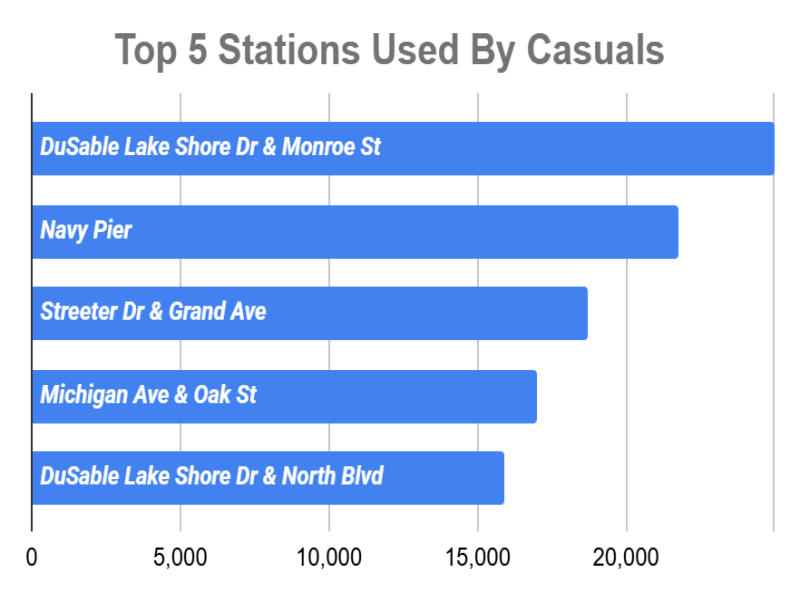


- **Average Trip Durations:** ***Casual users*** consistently recorded ***longer average trip durations*** than members across every day of the week. Both categories, however, saw their longest average trip durations on the weekends.





- **Station Analysis (Top 5):**
  - The top 5 start stations were identical to the top 5 end stations for the overall dataset.
  - When segregated, the top 5 start stations for members were also their top 5 end stations, and the same was true for casual users.



- Crucially, 4 of the overall top 5 stations were used mainly by casual users. This finding offers significant potential for targeted advertising, making the top 5 locations used exclusively by casual users a primary focus for the marketing team's strategies.

Summary Table:

Feature	Annual Members	Casual Riders
Primary Use	Work Commuting	Leisure/Recreation
Trip Start Hours	Bi-modal (Morning & Evening peaks)	Uni-modal (Evening peak)
Weekly Usage	Consistently high on weekdays; drops sharply on weekends	Low on weekdays; rises spontaneously, peaking on Saturdays
Average Trip Duration	Shorter average trip durations overall	Consistently longer average trip

		durations overall
Station Use	Top 5 start stations are also their top 5 end stations (likely commute hubs)	Top 5 start stations are also their top 5 end stations (likely leisure/tourist spots)
Bike Preference	High preference for Electric Bikes (63.5%)	Higher preference for Electric Bikes (66.8%)

Recommendations:

- ★ Casual riders are primarily leisure users who ride on weekends for longer durations. They may be motivated to purchase a membership if the value proposition addresses their specific use case, such as:
  - ➢ **Cost Savings for Extended Weekend Use:** Since they ride longer, a membership could offer a better rate for these extended trips compared to full-day passes.
  - ➢ **Convenience at Key Locations:** Targeted incentives at their highly-frequented leisure/tourist stations could demonstrate the benefit of immediate access and the membership discount.
  - ➢ **Perceived "Belonging" to a Community:** Digital media campaigns focusing on the social and lifestyle aspects of membership could appeal to their leisure-focused mindset.
- ★ Digital media allows for highly targeted, context-specific marketing that aligns with the identified usage patterns:
  - ➢ **Geotargeting:** Using the top 5 casual-user stations for digital ads (social media, mobile ads) within a defined radius.
  - ➢ **Timing:** Running high-impact campaigns as we approach weekends, when casual riders are most active.
  - ➢ **Messaging:** Creating digital content (videos, social posts) that highlights the benefits of membership for long, leisurely rides and weekend excursions.

Action Plan:

The marketing strategy should leverage the clear difference in usage patterns to target the leisure-focused casual riders.

1. **Targeted Station Advertising & Promotion:**
    - **Action:** Implement targeted digital and physical advertising at the **key stations** predominantly used by casual riders.
    - **Messaging Focus:** Highlight the **annual membership cost saving for long, weekend rides** (e.g., "Save $X on your long Saturday ride with a membership").
2. **Weekend-Focused Digital Campaigns:**
    - **Action:** Launch digital media campaigns (e.g., Google Display, Facebook/Instagram ads) that are heavily weighted toward **Saturdays and Sundays.**
    - **Messaging Focus:** Position the annual membership as the **"Weekend Pass Upgrade"** or "Leisure Rider's Choice," appealing directly to their higher weekend usage and longer average trip duration.
3. **Electric Bike Incentive Program:**
    - **Action:** Given the strong preference for electric bikes across both groups (especially casual riders), test a **limited-time digital promotion** that ties membership to electric bike use.
    - **Example:** *"Join now and receive a one-time bonus for your next 5 electric bike rides."* This capitalizes on a known preference to drive conversion.
4. **Content Marketing for Leisure:**
    - **Action:** Develop digital content (blog posts, social media maps, influencer collaborations) that maps out popular **leisure routes and weekend destinations** accessible via Cyclistic, showcasing the convenience and value of an annual membership for exploring the city.
    - **Goal:** Shift the perception of the membership from solely a commuter tool to a **leisure and lifestyle enhancement.**